$$\frac{\Delta g}{g} \times 100 = 4\%$$

The acceleration due to gravity on the earth's surface will approximately increase by 4%

Hence 4 is correct option.

Q. 6 The force required to stretch a wire of cross-section $1\ \text{cm}^2$ to double its length will be : (Given Yong's modulus of the wire $= 2 \times 10^{11}\ \text{N/m}^2$)

Option 1:
$1 \times 10^7\ \text{N}$

Option 2:
$1.5 \times 10^7\ \text{N}$

Option 3:
$2 \times 10^7\ \text{N}$

Option 4:
$2.5 \times 10^7\ \text{N}$

Correct Answer:
$2 \times 10^7\ \text{N}$

Solution:

$\text{A} = 10^{-4}\ \text{m}^2$

$\Delta\ell = \ell$

$\text{Y} = 2 \times 10^{11}$

$\text{Y} = \frac{\text{F}}{\text{A}} \times \frac{\ell}{\Delta\ell}$

$\text{F} = \text{AY}$

$= 2 \times 10^7\ \text{N}$

Hence 3 is correct option.

Q. 7 A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by 40°C, its efficiency increases by 30%. The temperature of the source will be :

Option 1:
$166.7\ \mathrm{K}$

Option 2:
$255.1\ \mathrm{K}$

Option 3:
$266.7\ \mathrm{K}$

Option 4:
$367.7\ \mathrm{K}$

Correct Answer:
$266.7\ \mathrm{K}$

Solution:

$$\eta = 0.5 = 1 - \frac{\mathrm{T_1}}{\mathrm{T_2}} \to 1$$

$$\eta' = 0.5 + \frac{30}{100}\,\eta = 0.65$$

$$\eta' = 0.65 = 1 - \left(\frac{\mathrm{T_1} - 40}{\mathrm{T_2}}\right)$$

$$0.65 = 1 - \frac{\mathrm{T_1}}{\mathrm{T_2}} + \frac{40}{\mathrm{T_2}}$$

$$0.15 = \frac{40}{\mathrm{T_2}}$$

$$\mathrm{T_2} = \frac{4000}{15} = \frac{800}{3}$$

$$\mathrm{T_2} = 266.7\ \mathrm{K}$$

The temperature of the source is $266.7\ \mathrm{K}$

Hence 3 is correct option.

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Q. 8 Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:

$\longrightarrow$ The average momentum of a molecule in a sample of an ideal gas is zero

$$\rightarrow V_{rms}=\sqrt{\frac{3RT}{M}}$$

$$\text{if } T_f=\frac{T_i}{2}$$

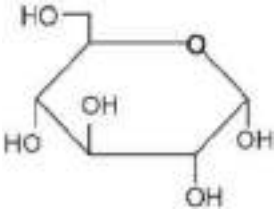
$$M_f=\frac{M_i}{2} \quad (\text{ after dissociation})$$

$$(V_{rms})_f=2\,(V_{rms})_i$$

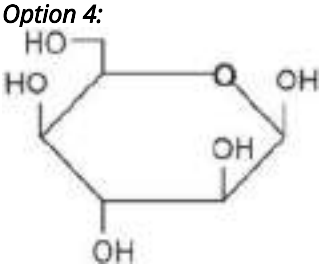
$\therefore$ Statement I is false but Statement II is true

Hence 4 is correct option.

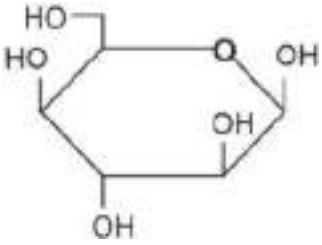
Option 3:



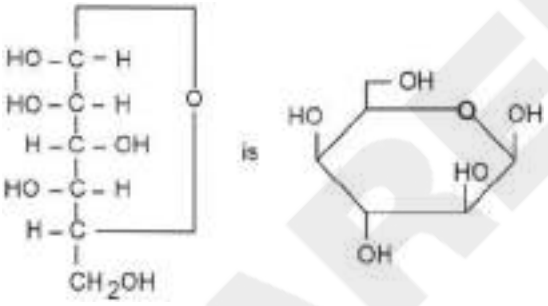
Option 4:



Correct Answer:



Solution:
The pyranose structure for -



Hence option (D) is correct.

Q. 19 Statements about Enzyme Inhibitor Drugs are given below :
(A) There are Competitive and Non-competitive inhibitor drugs.
(B) These can bind at the active sites and allosteric sites.
(C) Competitive Drugs are allosteric site blocking drugs.
(D) Non-competitive Drugs are active site blocking drugs.
Choose the correct answer from the options given below :

Option 1:
(A), (D) only

Option 2:
(A), (C) only

Option 3:
(A),(B) only

Option 4:
(A),(B),(C) only

Correct Answer:
(A),(B) only

Solution:
Enzyme inhibitor drugs are competitive and drugs which can bind at active and allosteric sites.

Hence option (C) is correct.

Q. 20 For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :
(A) Always use freshly prepared starch solution.
(B) Always keep the concentration of sodium thiosulphate solution less than that KI solution.
(C) Record the time immediately after the appearance of blue colour.
(D) Record the time immediately before the appearance of blue colour.
(E) Always keep the concentration of sodium thiosulphate solution more than that of KI solution.
Choose the correct answer from the options given below :

Option 1:
(A), (B), (C) only

Option 2:
(A),(D),(E) only

Option 3:
(D), (E) only

Option 4:
(A),(B),(E) only

Correct Answer:
(A), (B), (C) only

Solution:

Q. 30 The sum of all real values of x for which $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to _____.

Option 1:
No real values possible

Option 2:
0

Option 3:
2

Option 4:
4

Correct Answer:
No real values possible

Solution:

$$\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2-5x+12}$$

$$\frac{x^2+3x+10+2x^2-12x+7}{x^2+3x+10}=\frac{3x^2+5x+12+2x^2-12x+7}{3x^2+5x+12}$$

$$1+\frac{2x^2-12x+7}{x^2+3x+10}=1+\frac{2x^2-12x+7}{3x^2+5x+12}$$

$$(2x^2-12x+7)\left(\frac{1}{x^2+3x+10}-\frac{1}{3x^2+5x+12}\right)=0$$

$2x^2-12x+7=0$ or $3x^2+5x+12=x^2+3x+10$

$x=\frac{12\pm\sqrt{2}}{4}$ $\qquad 2x^2+2x+2=0$

$x^2+x+1=0$

Sum of Roots=6 $\qquad$ No solution

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28 July - Evening Shift

Physics

Q. 1 Consider the efficiency of carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta}\log e\frac{\beta x}{kT}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option

Option 1:
Dimensions of β is same as that of force.

Option 2:
Dimensions of $\alpha^{-1}\mathrm{x}$ is same as that of energy.

Option 3:
Dimensions of $\eta^{-1}\sin\theta$ is same as that of $\alpha\beta$.

Option 4:
Dimensions of α is same as that of β.

Correct Answer:
Dimensions of β is same as that of force.

Solution:

$$\eta = \frac{\alpha\beta}{\sin\theta}\log_e\left(\frac{\beta\mathrm{x}}{\mathrm{kT}}\right)$$

$$[\mathrm{KT}] = [\text{ Energy }] = \left[\mathrm{M}^1\mathrm{L}^2\mathrm{T}^{-2}\right]$$

$$\left[\frac{\beta\mathrm{x}}{\mathrm{KT}}\right] = [\mathrm{M}^\circ\mathrm{L}^\circ\mathrm{T}^\circ]$$

$$\therefore [\beta\mathrm{x}] = \left[\mathrm{M}^1\mathrm{L}^2\mathrm{T}^{-2}\right]$$

$$\rightarrow [\beta] = \left[\mathrm{M}^1\mathrm{L}^1\mathrm{T}^{-2}\right]$$

$$\because [\mathrm{x}] = \left[\mathrm{L}^1\right]$$